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DASStatFormer:

A Hybrid MultibranchTransformer with Statistical Feature Integration for DAS-Based Pattern Recognitions

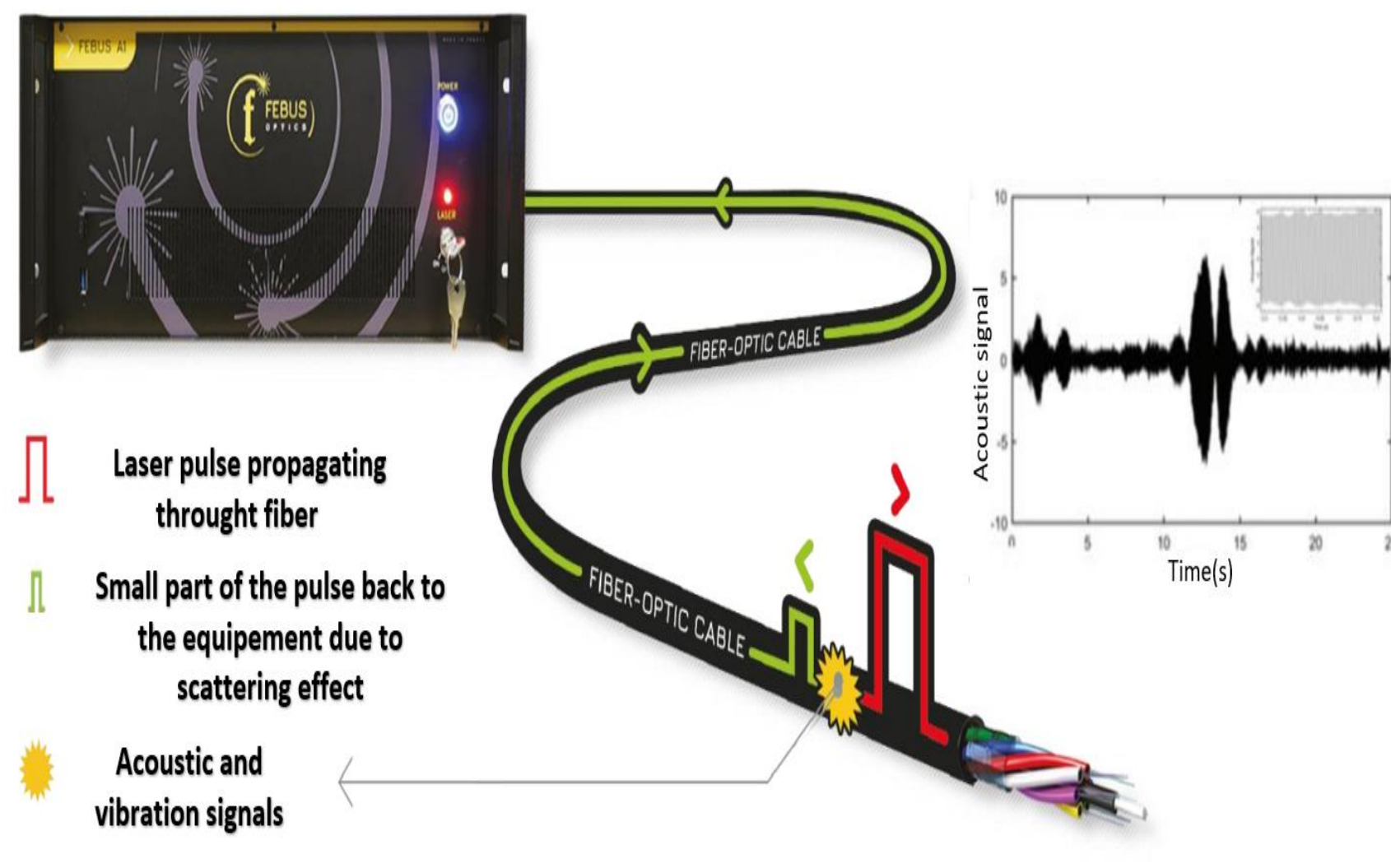
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1 Context & Problem Statement

Distributed Acoustic Sensing (DAS) enables large-scale monitoring via optical fibers, but its high dimensionality makes event classification costly: existing CNNs and Transformers either fail to capture long-range dependencies or require processing raw matrices at prohibitive cost.



$$X = \begin{bmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,n} \\ X_{2,1} & X_{2,2} & \cdots & X_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{m,1} & X_{m,2} & \cdots & X_{m,n} \end{bmatrix}$$

Bottleneck: DAS spatio-temporal matrices are extremely large (high sampling rate $\times$ dense channels). Existing Transformers (DASFormer, DeepViT) process raw signals $\rightarrow$ prohibitive computational cost and difficulty capturing both fine-grained patterns and global trends simultaneously.

2 Contribution

- Hybrid multibranch Transformer architecture combining multidomain statistical features with a Gated Transformer Network (GTN).
- Compact, interpretable 24-attribute representation (temporal, waveform, spectral), selected via ANOVA.
- Unified preprocessing pipeline restructuring real DAS recordings into standardized windows.
- Extensive validation (lab + real-world): strong accuracy, efficiency, and real-time scalability.

3 Multidomain Feature Extraction

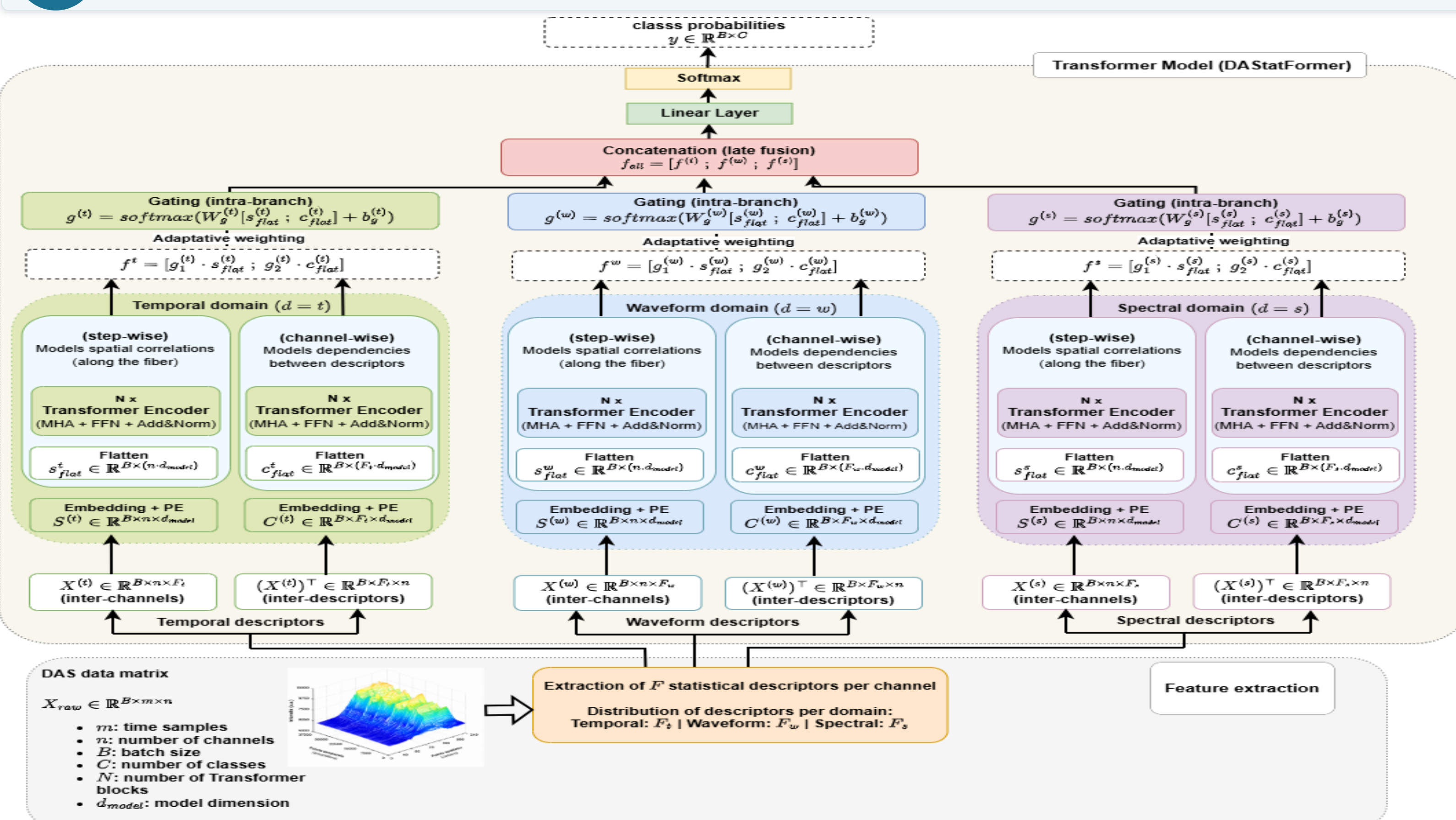
24 descriptor features per channel, ANOVA-selected from 60 candidates [1], spanning three complementary domains:

- Temp** Max amplitude, peak-to-peak, std. deviation, skewness, kurtosis, zero-crossing rate, envelope ratio
- Wave** Waveform, impulse, crest, and kurtosis factors; skewness / kurtosis of the Hilbert envelope
- Spec** Centroid, bandwidth, entropy, flatness, spectral slope, rolloff 85%/95%

Ex: $10,000 \times 12 \rightarrow 12 \times 24$

dimensionality reduction (raw segment $\rightarrow$ features/channel)

4 DASStatFormer Architecture



5 Configuration & Protocol

Model

Branches: 3 (time / waveform / spectral)
Encoders: N = 8 blocks / path
Model width: d_model = 256
FFN hidden: d_hidden = 128
Attention heads: h = 8
Dropout: 0.2 (MHA / FFN)

Training

Loss: Cross-Entropy
Optimizer: Adam, LR = 1×10^{-4}
Batch / epochs: 32 / 100
Gradient clip: $\|g\|_2 \leq 1.0$
Validation split: 10% (random)
Hardware: GPU CUDA:0

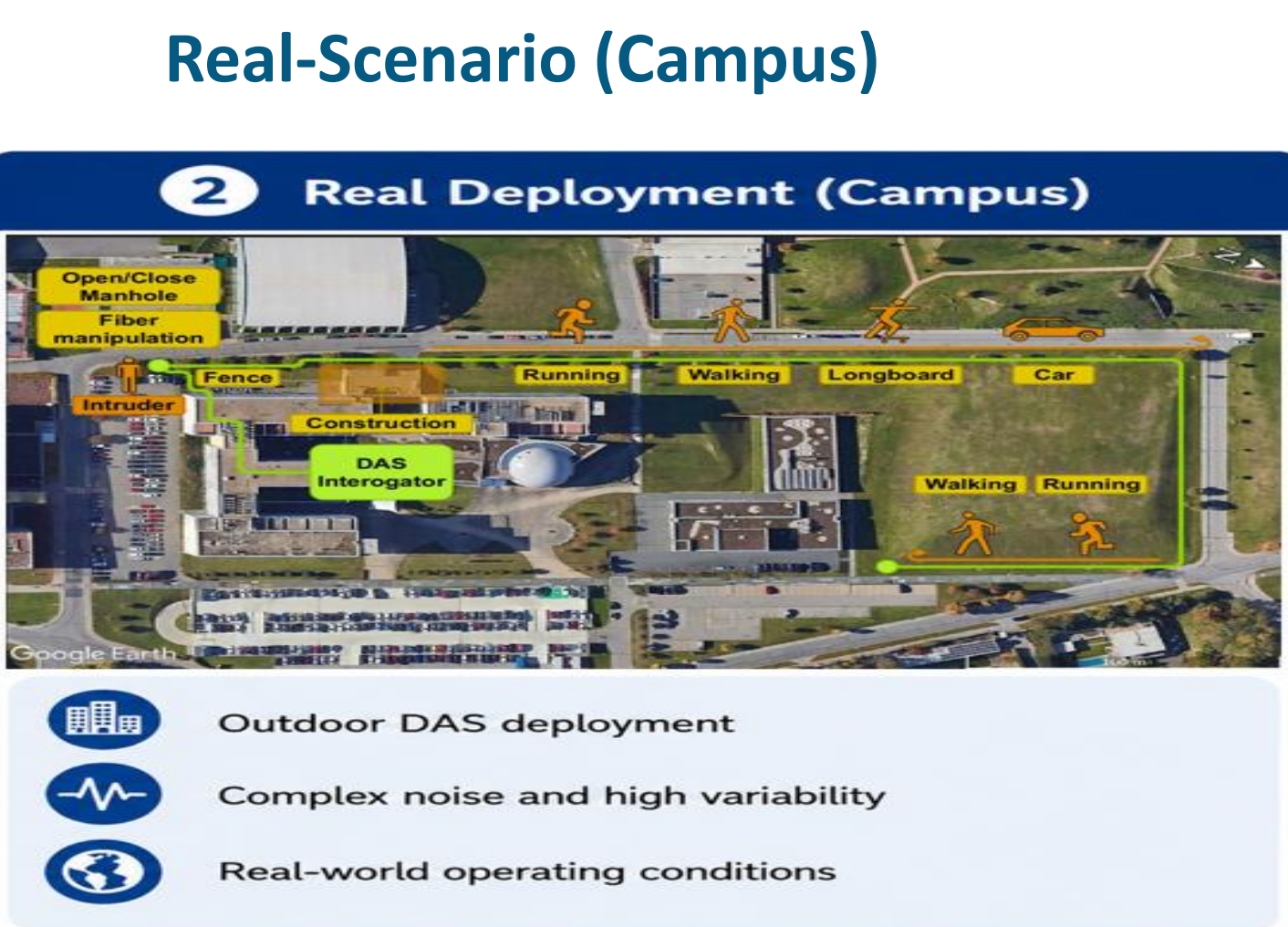
6 Datasets



15,612

(10000, 12) windows

Background, digging, knocking, watering, shaking, walking
Cao et al. [2]

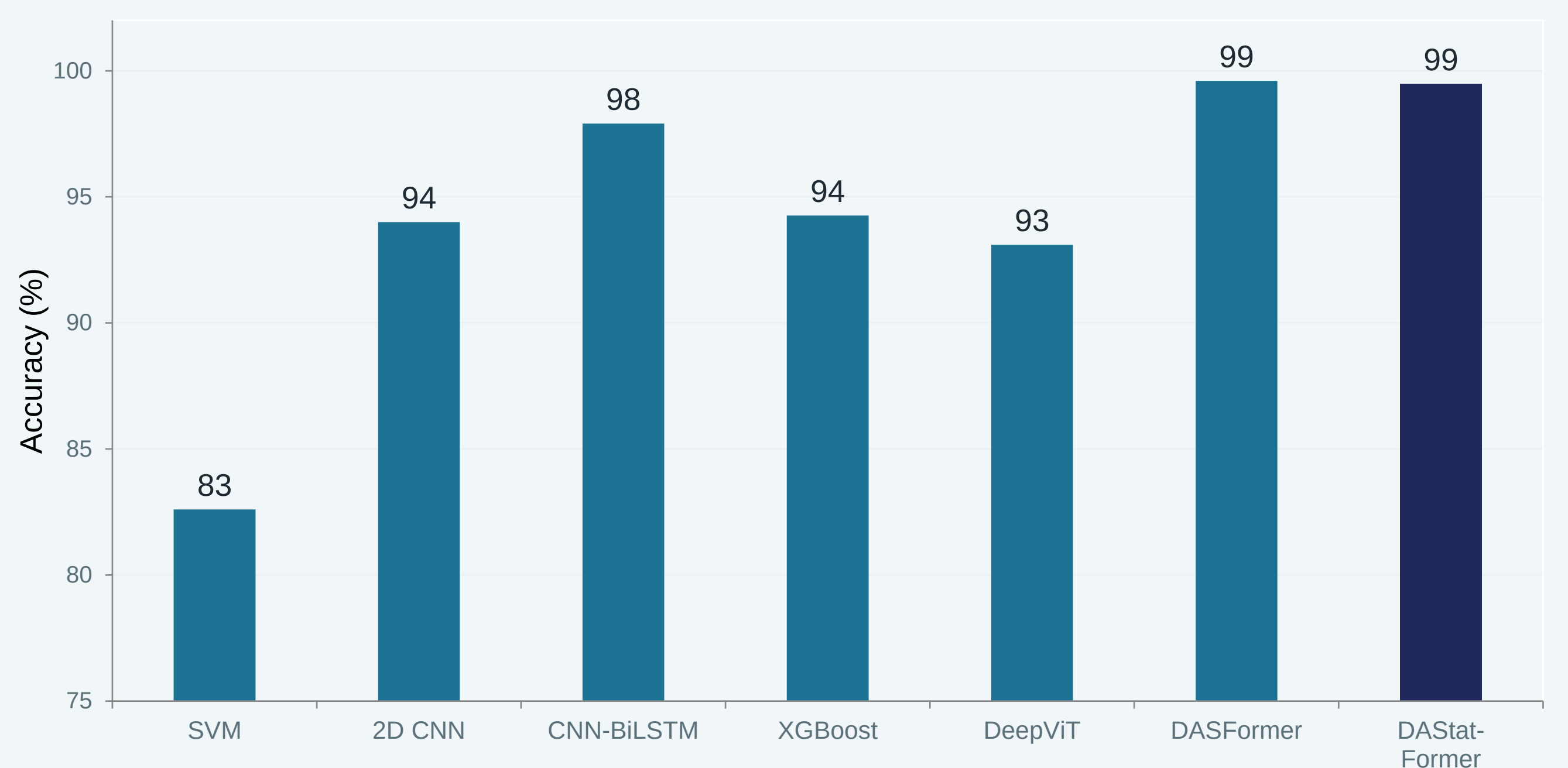


6,691

(2053, 1700) windows

Walking, running, longboarding, car traffic, fence climbing, fiber manipulation...
Tomasov et al. [3]

7 Results - phi-OTDR (Lab)



8 Results - Real-World Data

Model	Acc.	F1	Inf.(ms)	NAR
CNN [3]	91.4	86.8	—	—
DASViT1D	93.5	91.29	13	0.005
DASStatFormer (ours)	99.93	99.96	4.13	0.000

99.93%

accuracy (real-world)

4.13 ms

per sample

vs 13 ms for DASViT1D - simultaneous gains in accuracy and speed

9 Computational Efficiency

$\times 10$

fewer parameters vs DASFormer

115 MB

GPU memory allocated

$\rightarrow$ training and inference cost reduced by an order of magnitude, with no loss in accuracy

10 Conclusion & Future Work

DASStatFormer matches or exceeds existing CNN, hybrid, and Transformer-based approaches (>99.4% accuracy, near-zero NAR), while reducing computational cost by an order of magnitude — a solution well suited to large-scale, real-time DAS monitoring.

Future work: multi-sensor configurations • self-supervised learning • lightweight implementations for edge computing

[1] Hibert, et al.: Automatic identification of rockfalls and volcano-tectonic earthquakes at the Piton de la Fournaise volcano using a Random Forest algorithm. Journal of Volcanology and Geothermal Research (2017)

[2] Cao, X., Su, Y., Jin, Z., Yu, K.: An open dataset of phi-OTDR events with two classification models as baselines. Results in Optics 10, 100372 (2023)

[3] Tomasov, A., Zaviska, P., Dejdard, P., Klicnik, O., Horvath, T., Munster, P.: Comprehensive Dataset for Event Classification Using Distributed Acoustic Sensing (DAS) Systems. figshare (2025)